

EDUCATION

- **(Top 50) Monash University** Melbourne, Australia
PhD Degree (Supervised by Prof. Aldeida Aleti, Prof. Chunyang Chen, Prof. Hongyu Zhang) Oct 2022 – Aug 2026
 - Visiting Scholar: NLP Lab @ THU (Oct 2024 – Jun 2025), SE&AI Lab @ TUM (Jun 2025 – Aug 2025)
- **(Top 100) KTH Royal Institute of Technology** Stockholm, Sweden
Master's Degree in Machine Learning (Thesis Advisor: Prof. Martin Monperrus) Aug 2018 – Aug 2020
- **(Project 985) Shandong University (the Elite Program)** Jinan, China
Bachelor's Degree in Computer Science (Thesis Advisor: Prof. Jun Ma) Sep 2012 – Jun 2016

SELECTED PAPERS

- **Lexical Lens: Semantic Reference and State Transition in Language Models**
Jian Gu, Hongyu Zhang, Chunyang Chen, Aldeida Aleti Under Review
- **SemRF: A Semantic Reference Frame for Residual-Stream Dynamics in LMs**
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang Under Review
- **Rethinking Weight Tying: Pseudo-Inverse Tying for LM Stable Training and Updates**
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang Under Review
- **Cross-Scale Knowledge Transfer in LMs through Latent Semantic Alignment** ACL'26
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang (CORE A*)
- **Semantic-based Optimization for Repairing LLMs: Case Study on Code Generation** ICSE'26
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang (CORE A*)
- **Semantic-Aware Layer-Freezing for Computation-Efficient Fine-Tuning of LMs** ACL'25
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang (CORE A*)
- **Vocabulary-Defined Semantics: Latent Space Clustering for Beyond-Context Learning**
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang Under Review
- **Neuron Patching: Semantic-based Neuron-level LM Repair for Code Generation** TOSEM 2026
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang (JCR Q1)
- **Towards Top-Down Automated Development in Limited Scopes: A Neuro-Symbolic Framework from Expressibles to Executables** FSE'23 (IVR)
Jian Gu, Harald C. Gall (CORE A*)
- **Assemble Foundation Models for Automatic Code Summarization** SANER'22
Jian Gu, Pasquale Salza, Harald C. Gall (CORE A)
- **Multimodal Representation for Neural Code Search** ICSME'21
Jian Gu, Zimin Chen, Martin Monperrus (CORE A)
- **Source-Free Detection and Impact Analysis of Compiler Optimization in Mobile Apps** ASE 2026
Han Hu, Xiaoheng Xie, Bo Sun, Jian Gu, Gang Fan, Li Li (CORE A*)
- **On the Effectiveness of Transfer Learning for Code Search** TSE 2022
Pasquale Salza, Christoph Schwizer, Jian Gu, Harald C. Gall (JCR Q1)
- **Automated Classification of Overfitting Patches with Statical Code Features** TSE 2021
He Ye, Jian Gu, Matias Martinez, Thomas Durieux, Martin Monperrus (JCR Q1)

WORK EXPERIENCE

- **Ant Group** Hang Zhou, China
Research Intern Feb 2026 – Aug 2026
 - **LM Distillation**: proposed a novel distillation technique for LMs, enabling better support for edge-device tasks.
- **Huawei Research Center** Hong Kong, China
Research Intern Sep 2025 – Jan 2026
 - **Byte-Level Encoder**: compiler provenance recovery for performance load reduction in HarmonyOS ecosystem.
 - **On-Device Small LMs**: Small LM pretraining and post-training for intelligent coding in HarmonyOS IDE.
- **Monash University** Melbourne, Australia
Teaching Assistant and Co-Supervisor Jul 2023 – Jun 2024
 - **Teaching**: FIT5003 Software Security (1 semester); FIT4701/FIT4702 Projects and Research Theses (1 year).
- **University of Zurich** Zurich, Switzerland
Research Assistant, PhD Candidate Sep 2020 – Sep 2022
 - **Research**: Conducted research on AI for SE, including code retrieval, comment generation, and code generation.
 - **Teaching**: Designed and advised “AI Pair Programming”, a Master’s team project on code intelligence (1 year).
- **KTH Royal Institute of Technology** Stockholm, Sweden
Research Engineer Jun 2019 – Aug 2019
 - **Eclipse Repairnator**: Repairnator is a pioneering bot for automated program repair, hosted by the Eclipse Foundation. Integrated *Coming* and *SequenceR* into it and implemented its official GitHub app. Implemented *Prophet4J* (incl. *Sketch4Repair*) to generate code patches. Implemented *Context2Name* to prettify test cases.
- **Smart Software Studio** Yancheng, China
Software Engineer Jul 2017 – Jun 2018
 - **YoYo**: Delivered a technical solution including websites and apps for online community, with the supports for local advertisements. It is for identity-based socials between alumni and neighbors, serving over 2k human users.
 - **WiiPM**: Developed a 2D rogue-like game on the mobile platform. WiiPM contains elements of turn-based battles as well as collection and role-plays adventure. The techniques of procedural content generation are adopted.
- **UCar Inc.** Beijing, China
Software Engineer Mar 2016 – Nov 2016
 - **Anomaly Detection for Anti-Fraud**: Analyzed unusual operational logs, summarized anomaly patterns.
 - **Streaming Data Pipeline and Panel**: Maintained the streaming data pipeline for real-time transactions.
- **Hisense Group (R&D Center)** Qingdao, China
Algorithm Intern Sep 2015 – Feb 2016
 - **Movie GenePool**: Improved collaborative filtering recommender systems. Built an unified knowledge base using plot/review tags to fuse data providers. Developed a workflow for movie metadata processing and demonstration.

SERVICES

- **Software Engineering** ACM TOSEM, IEEE TSE (Reviewer)
- **Machine Learning** AAAI (PC Member); ARR, ICLR, NIPS (Reviewer)
- **Data/Info Management** CIKM (PC Member)
- **Multidisciplinary** WWW (PC Member); IEEE TAI (Reviewer)

HONORS & GRANTS

- **Award for Outstanding Overseas Study Elite** Chinese Government 2025 – 2026
- **Scholarship, Top-Ups & Supplementary Funding** Monash University 2022 – 2026
- **\$80k (2000 kSU) Research Grant** AU National Computational Infrastructure 2024 – 2026
- **\$30k HPC Research Grant** DUG Technology Ltd 2023 – 2024
- **Ranked #1 (until Jun 2020)** GitHub CodeSearchNet Challenge 2019 – 2020